Tom Hess

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PRODUCT-FOCUSED, AI-ENABLED ENGINEERING LEADER & SOFTWARE ENGINEER

Software engineer and engineering leader with 15+ years of experience building automation, integrations, and AI-enabled systems for SaaS and enterprise businesses, including two Director-level engineering roles. Hands-on work includes Java gift card and call-routing systems and RPA-driven billing automation. Since 2025, I have shipped apps and built an agent pipeline using an AI coding agent. Three years in product roles added a product lens, so I prioritize engineering work by customer outcome.

TECHNICAL SKILLS

Engineering & Integration: API / ERP / EDI Integrations | SaaS Platform Modernization | Root Cause Analysis | Technical Debt Management | SQL & Data Analysis | Process Mapping & System Diagramming

AI & Automation: Agentic AI Systems Design & Orchestration | AI-Assisted Delivery & Workflow Automation | AI-Assisted Application Development (Python, TypeScript, Rust) | Claude Code

Leadership & Delivery: Team Development & Coaching | Cross-Functional Leadership | Offshore / Distributed Team Management | Vendor Negotiation | Scrum and SAFe practice | Stakeholder Communication

Tools & Platforms: Jira | Confluence | GitLab | Jenkins | AWS | SQL Server | Power BI | Figma

TEAM & LEADERSHIP RECOGNITION

Tech Team of the Year – Cincinnati Business Courier (2019) | MRI Impact Award – Financial Modeling & Workflow Innovation (2018) | Corporate Tech & Innovation Award – Enterprise Automation (2017) | Realcomm Digie Award – Automation Leadership (2017)

EXPERIENCE

Raptor Technologies, Cincinnati, OH
Senior Product Owner, October 2023 – October 2026
(Acquired PublicSchoolWORKS, May 2024; Senior Product Manager until Oct 2025, then Senior Product Owner)

- Championed AI-DLC adoption, replacing manual Scrum story creation with a structured Intent-Elaborate-Build workflow, cutting planning overhead and accelerating delivery.
- Established team AI governance: approved AI tools and MCP servers, AI-specific logins for data protection, and a merge gate pairing automated AI review with a final human check on all AI-generated code.
- Led product development for compliance-driven LMS and workflow automation platforms serving ~45,000 schools nationwide, with ~99.9% customer retention.
- Eliminated regressions within 90 days through structured peer review, QA pipelines, and domain ownership, reducing support tickets by 20–30% and generating unsolicited positive customer feedback.
- Oversaw two delivery teams post-acquisition: an internal team of 3 engineers, 3 QA, and 1 DevOps lead, plus a vendor partner team of 1 PM, 3 engineers, and shared QA.
- Delivered Vehicle Accident Management, a new product, working closely with customers through definition and a pilot program; it drew immediate sales after reaching production.
- Coordinated phased Raptor–PSW SSO integration across three teams, delivering flat-file near-term value while laying groundwork for real-time API-based single sign-on.

Octus (formerly Reorg), Cincinnati, OH
Director of Engineering, July 2022 – September 2023

- Led root cause analysis on database and scalability outages, meeting 24–48 hour resolution SLAs on major incidents, where SLAs had previously been rarely met.
- Directed three cross-functional delivery teams with $500K budget oversight: 10 direct engineering reports plus two additional teams managed through an engineering manager.

- Implemented Scrum practices and documentation standards (runbooks, design docs, onboarding guides), reducing engineer onboarding time by 50%.
- Interviewed and hired engineers, and ran formal performance reviews for 10 direct reports.

Cengage Group, Cincinnati, OH
Manager, Software Engineering, January 2020 – July 2022

- Led distributed onshore and offshore engineering teams (10–12 engineers) supporting Cengage Unlimited, the flagship subscription learning platform serving millions of students.
- Delivered a RESTful Authorization & Permissions API (first touchpoint for every login) with sub-millisecond latency and 99.9% uptime by consolidating 5 legacy services.
- Retired 3 legacy applications through an SOA modernization project, collaborating with 20+ system owners to reduce complexity, maintenance burden, and costs.
- Increased sprint velocity by 50% through SAFe Agile practices, cross-team alignment, and coaching engineers into expanded roles.
- Reduced vendor costs from $40K to $1.2K annually through geolocation vendor consolidation, saving ~$39K a year across teams.

Phillips Edison & Company, Cincinnati, OH
Director / Associate VP, Enterprise Applications, October 2013 – January 2020
(Progressed from Application Developer → Manager → Director → Associate VP over 6+ years)

- Built Tenant Billing Automation Platform integrating ERP, vendor APIs (Billtrust), and RPA, cutting a 2-week billing cycle by 50–60% and freeing billing staff for direct tenant service and other higher-value work.
- Developed the Lease Profitability App alongside Finance and Leasing agents, cutting deal analysis from hours to seconds and giving agents real-time deal evaluation.
- Grew team from two developers to 12 engineers, contractors, and interns, training them on MRI, a real estate ERP, while overseeing a ~$1M annual budget.
- Built ICSC conference kiosk application delivering live leasing and availability data to agents during tenant negotiations, reducing deal close time by 1–2 days.
- Built SQL queries, stored procedures, and dashboards in the ERP and Excel cubes, letting Finance and Leasing staff make decisions from the data.
- Built a SQL Server training plan from Microsoft certification coursework into each engineer's growth path, reaching 90% completion and a ~10% YoY reduction in support bugs over 4 years.

Sterling Jewelers
Senior Programming Analyst, November 2010 – October 2013

- Built a nationwide gift card processing application in Java for Sterling Jewelers' 13 brands, eliminating external vendor reliance and bringing processing fully in-house.
- Built an automated IVR/call tree system in Java with real-time API integrations, reducing average customer call handling time by 25%.
- Unified the call-center agent desktop across CRM, external payment vendors, and credit account systems with IVR-triggered automation, reducing manual system-switching for agents supporting 13 retail brands.

Barnes Distribution
Programmer Analyst II → III, January 2009 – November 2010

- Led accounts payable side of the EDI payments project, automating JD Edwards data-file download and import steps and cutting check printing from 2 days to a couple of hours.
- Built VB.NET desktop and handheld-device applications for sales agents servicing clients, with the two communicating for ordering and custom reporting.

KeyBank
Application Systems Programmer, July 2007 – December 2008

- Developed and supported high-volume financial transaction systems and SOAP/XML APIs for ATM and banking operations on a team of about 10 engineers.

The Ohio State University, Columbus, OH
Research & Teaching Assistant, September 2005 – June 2007

- Completed M.S. research at the Center for Automotive Research on simulating a hybrid vehicle from software components.
- Taught introductory Java programming and software component modeling for two academic years.

SELECTED ENGINEERING PROJECTS

Independent Software Projects
Personal AI & Application Development, 2025 – Present
(Personal projects, built with an AI-enabled coding agent • Crosscue • CableHop • Retirement Planner)

- Built a 7-stage, local-first LLM agent pipeline with phase gates, then retired it for a leaner hosted harness when usage data showed guarded review wasn't worth maintaining.
- Shipped Crosscue, a Flutter crossword app, to the iOS App Store and Google Play, with automated daily puzzle retrieval and an account-free leaderboard on Cloudflare Workers and D1.
- Shipped CableHop to the Mac App Store, a Swift macOS app with a custom C USB/MTP engine, for managing Android devices over MTP.
- Rewrote a vanilla JS retirement planner as a React app, running the Monte Carlo simulation in a Web Worker off the main thread, with Zod-validated forms and reducer-based state.

EDUCATION

M.S.
Computer Science & Engineering
The Ohio State University
Thesis: Simulation of a Hybrid Vehicle Based on Software Components

B.S.
Computer Science & Engineering
The Ohio State University

CERTIFICATIONS

ITIL Foundation Certificate | Certified BMC Remedy Administrator | Claude Code 101 | Google Cloud Agent Fundamentals

TECHNICAL COACHING & APPLIED EXPERIENCE

FIRST Tech Challenge (FTC)
Robotics Head Coach – Software Development, 2023 – Present

- Coached teams to two World Championship qualifications (2023–24, 2024–25), including a top-4 finish at the Ohio State Championship and the software-specific FTC Control Award in 2024–25.
- Introduced the student team to progressively deeper engineering: methods and GitHub collaboration, layered architecture from motors and sensors up, and now AI-DLC with human oversight.

COMMUNITY & VOLUNTEERING

Crayons to Computers – Supply Distribution for Underserved Students | Hamilton County Parks – Seasonal Cleanup & Restoration Volunteer